

MLU MATHS

Tuesday, 14 September 2021

Course Outline

- **Lecture 1: Basic Linear Algebra**
 - Data Structures, Vectors, and Matrices
 - Vector Operations and Vector Spaces
 - Vector Norms
 - K-Nearest Neighbors (KNN)
 - Dot Products
 - Linear Models
(binary classification, KNN & Decision Plane)
- **Lecture 2: Transformations & Decompositions**
 - Matrix Multiplication
 - Ordinary Least Squares (OLS)
 - Matrix Transformations
 - Basis, Change of Basis
 - Eigen Algebra
 - Principal Components Regression
(regression, OLS & PC regression)
- **Lecture 3: Probability and Statistics**
 - Hypothesis Testing
 - Random Variables
 - Probability Distributions (discrete, continuous)
(hypothesis testing, AB testing)
- **Lecture 4: Differential Calculus**
 - Derivatives
 - Gradient Descent
 - Multivariate Testing
 - Automatic Differentiation
(binary classification, Logistic Regression)
- **Lecture 5: Bayes & Information Theory**
 - Fraud Detection
 - Bayes Theorem
 - Naïve Bayes
 - Information Theory
(binary classification, Naïve Bayes)
- **Lecture 6: Final Project Presentations**
(Recommender System)

This course covers two groups of content

Linear Algebra concepts, Differential Calculus, Probability and Statistics
Predictive methods based on analytical linear algebra on matrices (OLS, PCR),
probability theory (Naive Bayes)

ARITHMETIC --> BRANCH OF MATHEMATICS THAT STUDIES NUMBERS AND THEIR OPERATIONS
ALGEBRA --> BRANCH OF MATHEMATICS THAT STUDIES NUMBERS AND ABSTRACT SYMBOLS
LINEAR ALGEBRA --> BRANCH OF MATHEMATICS THAT STUDIES VECTOR SPACES AND LINEAR TRANSFORMATIONS
CALCULUS --> BRANCH OF MATHEMATICS THAT STUDIES HOW EQUATIONS CHANGE BY INCREMENTING
THE VALUE OF THEIR VARIABLES IN A SPECIFIC RANGE SUCH AS AREA, VOLUME, PATH, etc.
DIFFERENTIAL CALCULUS --> how it changes with respect to the derivative
INTEGRAL CALCULUS --> How it changes with respect to the integral
INFINITESIMAL CALCULUS --> How it changes with respect to infinity

Why might a model converge or not

When trying to Maximize the probability function with Gradient Descent on the LOSS function, not finding the minimum point of the curve since the tangent lines found for computing the weights of the we keep jumping to both sides without finding it
Convergence is reached when the neural network error significantly with respect to the number of iterations mejoran.

Convergence is an important concept in learning because the training process can take a long time and it is not always clear whether the network has learned enough
converged when training and validation error stops decreasing
optimal solution, this depends on many factors such as data quality, network architecture, and
A model can converge to a local minimum or equilibrium point instead of a
global minimum, resulting in sub-optimal performance

Convergence does not always mean the network found the global minimum of the loss function — it could
Therefore, it is important to monitor training and validation losses
and ensure it is not overfitting or underfitting
training data and perform well on the training set but poorly with unseen
Some optimization algorithms may find a poor solution due to
poor initialization, hyperparameter choice, or lack of convergence of the loss function
these problems, techniques such as regularization, data augmentation, and early stopping can be used to prevent
overfitting and improve the model's generalization capabilities

What does it mean for a model to be parametric or not and what are its practical implications
needs to go through all data points to make predictions
so it learns nothing and therefore its previous

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What is the difference between weights and bias?
Sometimes, a neural network can be overtrained and produce large uncontrollable weights that saturate neural signals too complex and leads to a machine learning concept called overfitting unnecessary data (or noise) that complicates the model's output prediction
Regularization reduces connection weights when performing These updates stabilize model generalization, i.e., its ability to adapt to new data

Biases are additional functions intended to help specify what a correct answer should look like
There can only be one bias per layer of neurons

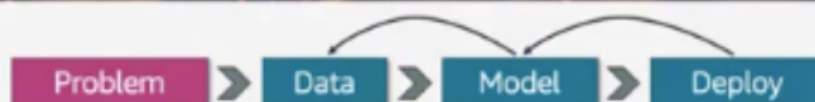
How does an additional matrix column differ from an additional dimension that turns it into a 3D tensor
3?

The difference is an additional value to one of its indices
The operations that can be performed on them are different
dimension of the vector
dimension of the matrix
Three or more indices $A_{i,j,k}$

Having completed this class, you should be able to:

- Understand the **fundamental math and statistics** used in machine learning
- Evaluate mathematical/statistical **approaches** to find the best way to **solve** different machine learning problems
- Use linear **algebra**, differential **calculus**, **probability** and **statistics** to solve machine learning problems
- Understand the essential mathematics of **gradient descent**

Math & Machine Learning



Topics in this course	Data	Model	Deploy
Basic Linear Algebra	Representing, Transforming, Operations on data at scale	KNN, Linear Classifiers	Compute big data faster
Intermediate Linear Algebra		PCA, <u>SVD</u> , Recommendation Engines	
Probability & Calculus	Randomness in data	Regression, Neural Networks, Gradient	